Temperature impacts fate of antibiotic resistance genes during vermicomposting of domestic excess activated sludge

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Abstract

Effect of temperature on antibiotic resistance genes (ARGs) during vermicomposting of domestic excess sludge remains poorly understood. Vermicomposting experiment with excess sludge was conducted at three different temperatures (15°C, 20°C, and 25°C) to investigate the fate of ARGs, bacterial community and their relationship in the process. The vermicomposting at 25°C did not significantly attenuate the targeted ARGs relative to that at 15°C and 20°C. The dynamics of *qnr*A, *qnr*S, and *tet*M genes during vermicomposting at 15°C and 20°C followed the first-order kinetic model. Temperature remarkably impacted bacterial diversity of the final products with the lowest Shannon index at 25°C. The presence of the genus (*Aeromonas* and *Chitinophagaceae*) at 25°C may contribute to the rebound of the genes (*qnr*A, *qnr*S and *tet*M). The study indicates that 20°C is a suitable vermicomposting temperature to simultaneously reach the highest removal efficiency of the ARGs and the good biostability of the final product.

Key word: emerging pollutants; earthworms; first-order kinetic model; sludge resources utilization; temperature variation

1 Introduction

Excess activated sludge, an inevitable by-product from domestic wastewater treatment plant, is enriched with large amounts of antibiotic resistance genes (ARGs) (Xu et al., 2015), and its potentially environmental risk is receiving the growing concerns in recent years. Excess sludge usually needs to be properly treated prior to entering into various soil environments to minimize the adverse effects of toxic substances such as heavy metal, pathogenic bacteria, antibiotics, and microplastics (Sun et al., 2019a; Huang et al., 2020; Li et al., 2020), on human health and the ecological environment.

Main biologically treatment technologies for management of excess sludge, include thermophilic composting and anaerobic digestion, which have been widely applied worldwide. Lots of studies have illustrated that both approaches would significantly affect the fate of ARGs in excess sludge. By using high-throughput quantitative PCR (polymerase chain reaction), Su et al. (2015) observed that composting could remarkably raise the abundance and diversity of most ARGs in the sewage sludge probably due to the occurrence of abundant ARGs host bacteria (e.g. actinobacteria), whilst hyperthermophilic (the highest temperature up to 90°C without exogenous heating) composting is regarded as an efficient way for the removal of ARGs in the sewage sludge (Liao et al., 2017). On the other hand, Cui et al. (2020) reviewed and confirmed the positive effects of anaerobic digestion on ARGs elimination in various biowastes and at different temperatures. It is generally accepted that the fate of ARGs in the sludge varied depending on the selected treatment methods and operating conditions (e.g., temperature and hydraulic retention time) (Ma et al., 2011; Liao et al., 2017; Sun et al., 2019b), as well as the types of the detected resistance genes. However, the involved mechanisms in terms of ARGs attenuation during the process of treatments needs to be clarified.

Vermicomposting is regarded as a traditional and eco-friendly approach to transform organic solid wastes such as sewage sludge, agricultural waste and livestock manure, into nutrient-rich products by the joint action of earthworms and microorganisms (Bhat et al., 2017; Domínguez et al., 2021). Recent studies have suggested that vermicomposting treatment of sewage sludge enabled to effectively eliminate tetracyclines and quinolone resistance genes (Huang et al.,2018; Cui et al., 2018), and high inoculation density of earthworms could also enhance the removal efficiency (Cui et al., 2018), which is possibly attributed to the gut microbes of earthworms (Cui et al., 2019). From the application perspective of vermicomposting technology, temperature is a more easily controlled environmental factor, and an appropriate increase in temperature contributed to the improvement of vermicompost stability reflected by dehydrogenase activity (Chen et al., 2016), and dissolved organic carbon (Zhang et al., 2020). Garg and Gupta (2011) observed that there was higher earthworm biomass in the vermicomposting system of household waste in winter season than in summer season, and the final vermicompost was more abundant in nutrients (N, P and K). Earthworm (*Eisenia fetida*) showed good adaptability at the range of 15°C~25°C (Edwards and Lofty, 1977). Temperature variations in vermicomposting would lead to different microbial profiles (activity and diversity) in the treated products (Zhang et al., 2020). Hou et al. (2021) found lower bacterial diversity and biostability in the final vermicompost at 25°C compared with that at 15°C and 20°C. Since ARGs and intermediates of their dissemination are closely linked to the bacterial populations during biological-treatment of organic wastes (Ma et al., 2011; Su et al., 2015), it could be speculated that ARGs would differently response during vermicomposting of excess sludge under varying temperature conditions and its fate is expected to have a close relationship with bacteria community. However, study on the effect of temperature on ARGs during vermicomposting of excess sludge is still lacking. A recent study regarding effect of biochar on ARGs during vermicomposting of sewage sludge, designed a relatedly wider temperature range (18°C-28°C) (Huang et al., 2020), which may underestimate the effects of temperature fluctuation on ARGs even though there was relatively higher efficiency for ARGs removal after adding biochar. The present study designed a vermicomposting system in a relatively suitable temperature range (15°C-25°C), which is expected to elucidate the influencing mechanisms of temperature variation on ARGs.

The aim of the present study was to gain an insight into the responds of ARGs to vermicomposting of domestic excess sludge at different temperatures. For this, a lab-scale vermicomposting of excess sludge was established at 15°C, 20°C and 25°C that represent suitable temperature range for earthworm's growth, respectively. Typical resistance genes including tetracycline resistance genes (*tet*G, *tet*M and *tet*X), sulfonamide resistance gene (*sul*1), quinolone resistance genes (*qnr*A and *qnr*S) as well as integrase gene (*intl*1) that is an indicator of horizontal propagation of resistance genes between bacterial cells were selected and quantified by quantitative PCR with the specific primers. Moreover, to clarify the response mechanism of the selected resistance genes to vermicomposting at different temperatures, bacterial community composition was examined by utilizing high throughput sequencing.

2 Materials and methods

2.1 Earthworm and excess activated sludge

Earthworm species of *Eisenia fetida* that is widely adopted in the vermicomposting system due to its strong tolerance to wide temperature range and unfavorable environments along with high vermicomposting efficiency was used for present study. Dewatered excess activated sludge was obtained from Qilihe-Anning wastewater treatment plant in Lanzhou City, China. Dewatered sludge was pelletized into granular sludge with a size of 5 mm referring to the method developed by Fu et al. (2015). Pelletization was confirmed to be capable of improving the aerobic conditions inside the sludge, and save simultaneously the addition of bulk materials (Fu et al., 2015). The physical, chemical and biological properties of the pelletized sludge used in the vermicomposting experiment are presented in **Table 1**.

2.2 Vermicomposting design

Hundred clitellum earthworms with approximately 1.0 g per individual were inoculated into each reactor with a size of 12 cm × 36 cm in height × diameter with the addition of 4 kg pelletized sludge. The reactors were covered with plastic films to keep a related unchanged moisture inside of the reactors and placed in incubators set at 15°C, 20°C, and 25°C, respectively, according to the previous research (Edwards et al., 1977) combined with our several studies (Chen et al., 2016; Fu et al., 2017). All treatments were triplicated. Vermicomposting was conducted for 60 days to ensure relatively good biostability of the final residues according to our previous study (Fu et al., 2016) and the sampling interval was 10 days. To ensure the same inoculation density (25 earthworms / kg wet sludge) of earthworms during the whole vermicomposting, 5 earthworms were randomly taken out at each sampling of sludge (approximately 200 g wet sample for each reactor). One part of the obtained samples was measured for water content, organic matter and dehydrogenase activity (DHA) in 48 hours. Another part of the sample was air-dried, ground and sieved with 100-mesh for physical-chemical properties. The remaining sample was frozen-dried and then stored at -40°C for ARGs and bacterial community analysis.

Table 1 Properties of initial pelletized dewatered sludge

Parameter	Initial pelletized dewatered sludge
Water content (%)	68.0±7.7
Organic matter (%)	80.4±0.01
Dissolved organic carbon (g/kg)	16.7±2.3
Electrical conductivity (µS/cm)	573±8.5
pH	6.77±0.11

Ammonium (mg-N/kg)	7.4±0.1
Nitrate (mg-N/kg)	10.3±2.0
Dehydrogenase activity (mg-TF/g/h)	28.1±1.3

Means ± errors (n=3).

2.3 Analysis of physical, chemical, and biological properties

Organic matter (OM) was determined by burning a sample in a muffle furnace at 600°C for 2 hours. Dissolved organic carbon (DOC) was measured according to oxidizing method with sulfuric acid-potassium dichromate (Cui et al., 2018). In detail, dry sample was dissolved in distilled water at a ratio of 1:10 (mass: volume), and the obtained liquid was filtered through a membrane with a pore size of 0.45 μm. After the filtration, the filtrate was oxidized with potassium sulfate-dichromate at 200~220°C and then titrated with ferrous sulfate solution. At the same time, the unfiltered mixed liquids above were used for the measurement of electrical conductivity (EC) with a conductivity meter. Ammonium nitrogen (NH_4^+-N) was analyzed by spectrophotometry after the sample was dissolved with potassium chloride solution. Nitrate nitrogen (NO_3^--N) was determined using colorimetric method according to the previous study (Fu et al., 2015). DHA was referred to the TTC reduction method reported by Fu et al. (2015).

2.4 Quantification of ARGs and bacterial community

(1) DNA extraction

Total genomic DNA from the frozen sample (approximately 0.3 g) was extracted by using the Power Soil DNA Extraction Kit (MOBIO, USA) according to the manual instructions. Bacterial abundance was analyzed by a fluorescent quantitative PCR instrument (Takara, Japan). Primer information used for amplification is referred to the previous study (Cui et al., 2018). 25 μL PCR reaction solution included 12.5 μL SYBR Green, 0.5 μL primers with a concentration of 10 μM, 2 μL DNA template and the rest was supplemented with sterilized water. Negative control without DNA template was also setup to exclude the sample pollution. Three technical replicates for all PCR were performed. PCR amplification condition consisted of pre-denaturation 95°C for 3 mins, 40 thermal cycles (95°C for 5 s, 60°C for 30 s). A bacterial quantitative PCR standard curve was made with known concentrations of standards of *E. coli*. The amplification efficiency of the standard curve for real-time PCR was stable in the range of 85-106%, implying that the quantitative analysis requirements were satisfied.

(2) Determination of ARGs and integrase gene by quantitative PCR

The targeted ARGs used in the present study included quinolone resistance gene (*qnr*A and *qnr*S), tetracycline resistance gene (*tet*G, *tet*M, and *tet*X) as well as sulfonamide resistance gene (*sul*1). The detailed information regarding the primers and protocols used in the quantitative PCR is the same as the previous study (Cui et al., 2018; Cui et al., 2019). The amplification efficiency of the standard curve for real-time PCR was stable in the range of 85-106%, implying that the quantitative analysis requirements were satisfied.

(3) Determination of bacterial community via high-throughput sequencing

The extracted DNA samples above were submitted to Shanghai Majorbio Bio Pharm Technology Co., Ltd for sequence analysis. In detail, PCR analysis was firstly performed with primer pair of 338F (5'-ACT CCT ACG GGA GCA GCA-3') - 806R (5'-GGA CTA CHV GGG TWT CTA ATT-3') (Chen et al., 2016), and then the library was constructed and sequenced through Illumina MiSeq 300 platform. Finally, the obtained sequence data was filtered for bioinformatic analysis including alpha diversity indices (OTU, chao, and Shannon) via QIIME software (Version 1.7.0) and shown with R software (Version 2.15.3). Beta diversity on weighted unifrac was calculated with QIIME software (Version 1.7.0). Sequencing data of the bacterial 16S rRNA genes were deposited in the National Microbiology Data Center (NMDC) database with accession no. NMDC40013966~NMDC40013984.

2.5 Data analysis

All data were normalized and homoscedasticity before ANOVA treatment with the SPSS software. One-way ANOVA with mean separation by Tukey's significant difference (HSD) test at a significance level of $p < 0.05$ was carried out to evaluate the pairwise comparison. Referring to a previous study (Liao et al., 2017), a first-order kinetic equation was also constructed to better gain insight into the dynamics of ARGs during vermicomposting of sludge at different temperatures. The relationship between the resistance genes and the bacterial community was also discussed by applying the redundancy analysis. Redundancy analysis (RDA) of relationship between resistance genes and physicochemical parameters as well as bacterial community was made by using Canoco 4.5. Pearson correlation between bacteria (genus level) and resistance genes as well as integrase gene was assessed by using SPSS software at a significance level of $p < 0.01$.

3 Results and Discussion

3.1 Dynamics of ARGs and *intl*1 gene under different temperature conditions during vermicomposting

The dynamics of absolute abundance of three types of resistance genes and integron gene in the vermicomposting process are shown in **Figure 1**. For the initial pelletized sludge, the abundance of sulfonamide and tetracycline resistance genes with an order of magnitude of E8-9 was higher than quinolone resistance genes with an order of magnitude of E5-7, whereas the integrase gene (*intl*1) had the highest absolute abundance with a magnitude of E10. The obtained findings indicated that the domestic excess sludge not only contained abundant ARGs, which is agreement with the previous studies, but also had high horizontal transfer potential for resistance genes (Yang et al., 2014; Sun et al., 2019b).

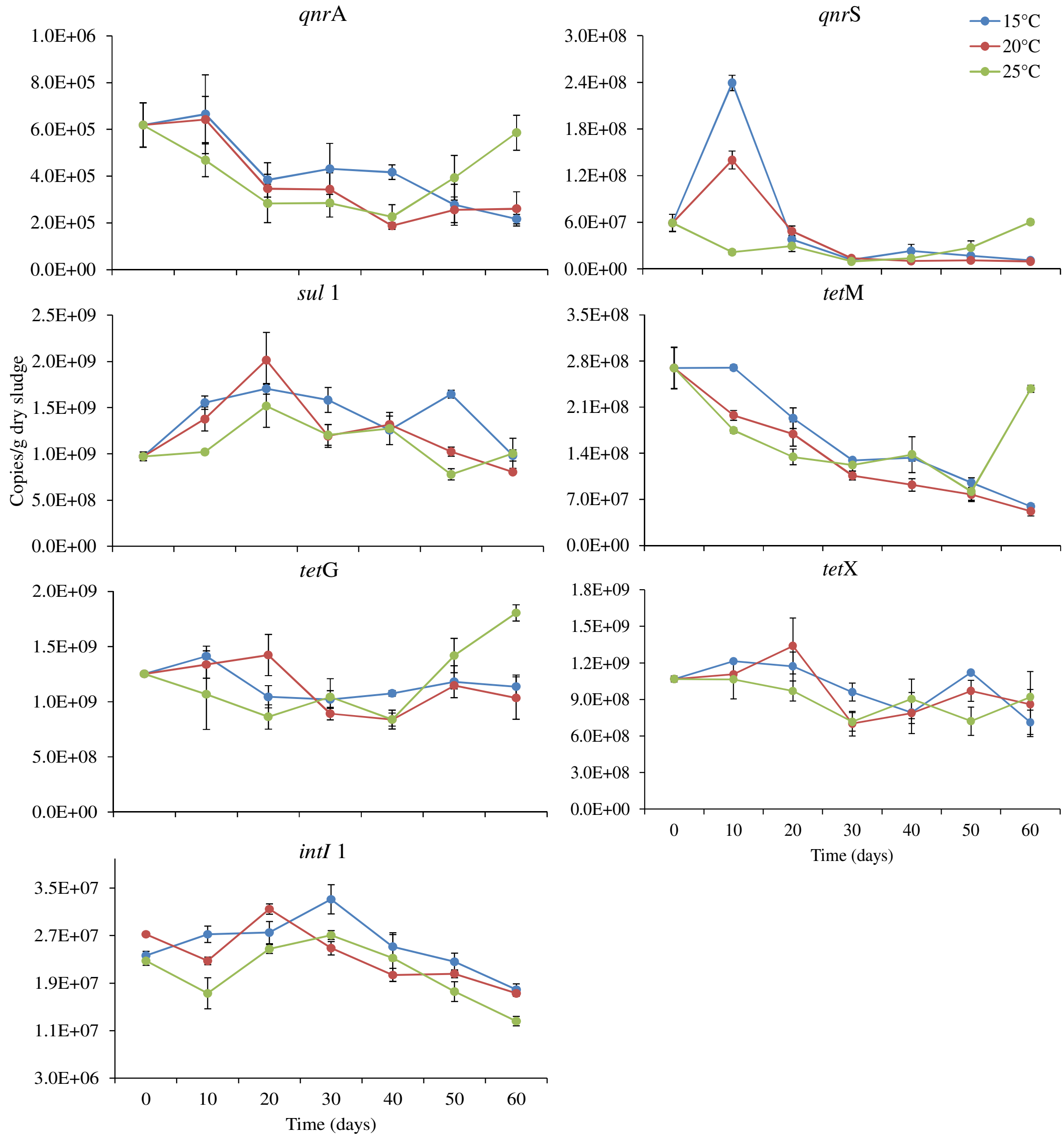

Figure 1 The absolute abundance of ARGs and *int*1 gene during vermicomposting of domestic excess sludge at 15°C, 20°C and 25°C

The effect of vermicomposting process on the resistance genes was closely associated with the type of selected resistance genes. The abundances of two quinolone resistance genes (*qnr*A and *qnr*S) were significantly declined during the whole vermicomposting process. However, this resistance gene rebounded at the end of vermicomposting at 25°C. The vermicomposting had not a significant effect on the *sul* gene, and the effect of temperature variation was not obvious. There was a remarkable decrease in *tet*M gene, but *tet*G and *tet*X genes did not present significant changes in the abundance, and they also rebounded in the later stage of vermicomposting at 25°C. Besides, vermicomposting at 25°C favored the reduction of *intl*1. Overall, vermicomposting at 25°C did not show significant attenuations in the abundance of resistance genes in comparison to that of 15°C and 20°C, which is not in coincidence with the stabilization result of vermicomposting. The reason behind this difference may be related to the discrepancy of the community structure of bacteria hosting the above-mentioned resistance genes under different temperature conditions. Zhang et al. (2020) and Hou et al. (2021) found significantly specific bacterial community in the final products of sludge vermicomposting at 25°C compared to 15°C and 20°C, may partly explain the discrepancy of ARGs caused by temperature variation. The following microbial community analysis in the present study would further illustrate the difference. Relative abundance (ARGs/16S rDNA) of resistance genes is directly associated with the dynamics of the host bacteria (Cui et al., 2019). Changes in the relative abundance of the resistance genes in the vermicomposting process are shown in **Figure 2**. *qnr*A*, qnr*S and *tet*M genes were significantly reduced in the relative abundance, whereas other genes did not significantly change, which suggests that vermicomposting had significant removal efficiency for some resistance genes, and also exhibited selectivity for the reduction of the three types of ARGs. It is probably related to a fact that free DNA existed in the excess sludge is more susceptible to be degraded compared to intracellular DNA (Cui et al., 2019), thereby the corresponding carried resistance genes would exhibit different attenuation characteristics.

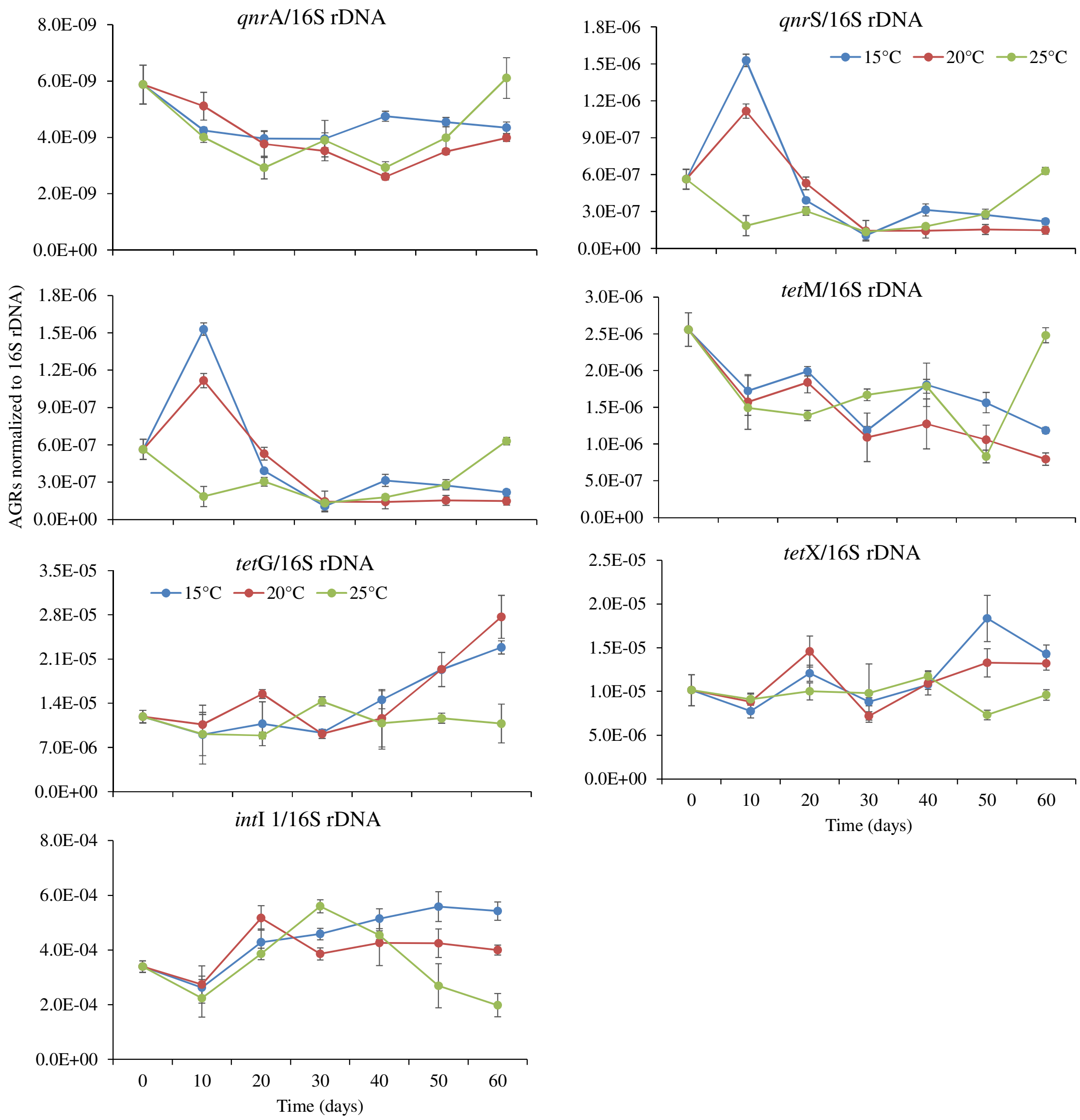

Figure 2 The relative abundance of ARGs and *intI*1 gene during vermicomposting of domestic excess sludge at 15°C, 20°C and 25°C.

The first-order kinetics for the three types of ARGs and integrase gene during vermicomposting were generated with coefficients and correlation as shown in **Table 2**. The dynamics of *qnr*A, *qnr*S, and *tet*M genes during vermicomposting at 15°C and 20°C fitted to the first-order kinetic equation and the kinetic coefficients at 20°C was higher than that at 15°C, which suggests the vermicomposting at 20°C had more significant removal of the targeted resistance genes. However, compared to vermicomposting, thermophilic composting especially hyperthermophilic composting and anaerobic digestion exhibited more effective removal for ARGs according to the results of kinetic coefficients (Liao et al., 2017; Diehl and LaPara, 2010). The combination of thermophilic composting and vermicomposting is a feasible treatment strategy to efficiently decrease the ARGs in sewage sludge and is recommended by many previous studies in terms of efficient treatment of biowastes (Fornes et al., 2012; Ravindran and Mnkeni, 2017; Ning et al., 2021).

Table 2 First-order kinetic model analysis of ARGs and *intl*1 during vermicomposting of domestic excess sludge.

Target genes	Kinetic Coefficients (k, d^{-1})			Correlation (r^2)		
	15°C	20°C	25°C	15°C	20°C	25°C
*intl*1	0.0047	0.0027	0.0057	0.2742	0.2588	0.2871
*qnr*A	0.0152*	0.0190*	0.0026	0.8393	0.7164	0.0217
*qnr*S	0.0271*	0.0330*	0.0008	0.5125	0.6990	0.0006
*sul*1	0.0006	0.0057	0.0022	0.0026	0.1726	0.0488
*tet*G	0.0025	0.0025	0.0016	0.1645	0.0381	0.0597
*tet*M	0.0220*	0.0269*	0.0067	0.9239	0.9834	0.1238
*tet*X	0.0037	0.0041	0.0054	0.3374	0.2387	0.3471

Asterisk (*) represents significant difference at the level of $p < 0.01$.

3.2 Dynamics of bacterial community under different temperature conditions during vermicomposting

In order to clarify the mechanism of variation of ARGs during vermicomposting of sludge under different temperature conditions, bacterial community including diversity and composition was investigated. Shannon diversity index in the vermicomposting process fell into the range of 4.78-5.91 (**Table S1**), and reached to 5.05 at 15°C, 5.91 at 20°C, and 4.78 at 25°C, suggesting that bacterial diversity was also affected by vermicomposting temperature. The lowest bacterial diversity observed in the final product at 25°C was probably ascribed to the highest bio-stability (Huang et al., 2020). Furthermore, the OTU presented the lowest value in the final vermicomposting product at 25°C and the highest value at 20°C, which is in line with Shannon index. These results underscore that changing vermicomposting temperature would impact bacterial community diversity of the final products.

Figure 3(a) shows the dynamics of bacterial community composition at the phylum level during the vermicomposting of sludge. Primary phylum included Proteobacteria, Bacteroidetes, Actinobacteria, Chloroflexi, Firmicutes, and Acidobacteria. Cluster analysis revealed that the samples taken at the 50 day and 60 day at 25°C, as well as 60 day at 20°C were affiliated into an independent group which is obviously different from the other samples, which indicate that the bacterial community composition at the end of vermicomposting especially at 20°C and 25°C greatly changed. In detail, the relative abundance of bacteria including Proteobacteria, Actinobacteria, Chloroflexi, and Firmicutes was observed to be lower at 25°C compared to that at 20°C, while the relative abundance of Bacteroides at 25°C was higher than that at 20°C. The reduced abundances of Proteobacteria and Firmicutes suggest that excess sludge may undergo more complete degradation under higher temperature conditions. Firmicutes belongs to a facultative anaerobic bacterium (Wang et al., 2016). Earthworm activity at 25°C is relatively higher, and its burrowing activity can enhance the oxygen level of substrates, hence leading to a lower abundance of Firmicutes. Bacteroidetes can degrade refractory organics such as chitin and cellulose (Chen et al., 2018), therefore, the observed highest abundance at 25°C indicates that the stability of vermicompost under this temperature condition was the best compared to that at 15°C and 20°C, which is well confirmed by biochemical properties of final products, such as DOC, NH_4^+/NO_3^- and DHA (**Figure. S2**). Furtherly, the highest biostability of the vermicompost at 25°C may imply different ARGs occurrence.

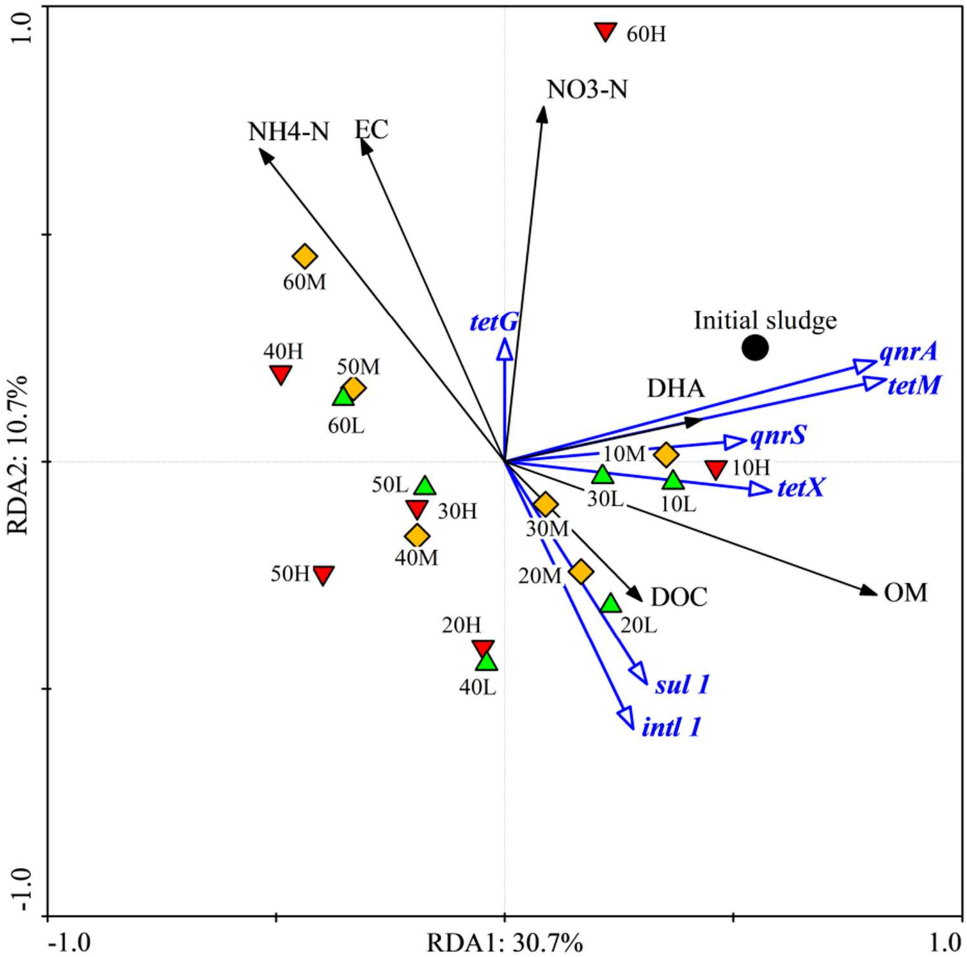

Figure 3 Redundancy analysis of relationship between ARGs and physicochemical properties. IS-initial sludge. L, M and H represent treatments at 15°C, 20°C and 25°C. The number at the front of L, M and H represent sampling time (day). OM-organic matter, EC-electrical conductivity, DOC-dissolved organic carbon, DHA, dehydrogenase activity.

In addition, dynamics of bacteria at the genus level during vermicomposting were also analyzed, as shown in **Figure 3(b)**. The bacteria are clustered into two groups, and of the genus, *Arenibacter*, *Halomonas* and *Niabella* were clustered together. Their abundances increased at the end of vermicomposting stage ranging from 0.00% to 0.41%, 0.16%, 10.68% for *Arenibacter*, from 0.02% to 0.18%, 0.21%, 1.67% for *Halomonas*, from 0.01% to 0.09%, 0.51%, 2.05% for *Niabella*. *Arenibacter*, a strictly aerobic bacteria, its optimum growth temperature is 25°C and can produce catalase and oxidase (Jeong et al., 2013). *Niabella*, a strictly aerobic bacteria, its optimum growth temperature is in the range of 25~30°C (Kim et al., 2007). The aerobic status in the final vermicomposting products and the optimal temperature condition were responsible for the remarkable increase in abundance of these bacteria. Moreover, *Zoogloea*, *Sediminibacterium*, *Trichococcus*, *Longilinea* and *Macellibacteroides* were remarkably decreased in abundance at 25°C in comparison to that at 15°C and 20°C. *Zoogloea* plays an important role in the formation of activated sludge flocs by absorbing nutrients from the environment. A significant decrease in abundance at 25°C indicates that the floc structure of the sludge was destroyed, which may favor further degradation of organic matter. *Sediminibacterium* is an organics-degrading bacterium under strict aerobic conditions, and its optimal growth temperature is in the range of 22-28°C (Qu and Yuan, 2008). Therefore, the genus was able to exert the maximum degradation ability for the sludge at 25°C. *Trichococcus* was found to be capable of transforming complex organics into small molecular compounds such as lactate, acetate, formate, and ethanol that can be readily utilized by other microorganisms (Song et al., 2018; Dai et al., 2017). Therefore, a significant decrease in the abundance of the bacteria at 25°C confirms that the excellent stability of domestic excess sludge treated via vermicomposting.

3.3 Correlation between ARGs and bio-stability parameters & bacterial community

Based on RDA analysis of relationship between sludge properties and ARGs involved during vermicomposting (**Figure 4**), parameters including OM, NH_4-N, NO_3-N and EC had significant effects on the ARGs and *intI*1gene. Of these, NH_4-N, NO_3-N and EC had significantly negative correlations with *sul*1 and *intI*1 genes. The difference caused by vermicomposting temperature started from day 10 until the end of the treatment, and the final product at 25°C notably differed from the treatments at 15°C and 20°C. A significant increase of nitrate-nitrogen at 25°C was probably the main reason for the rebound of *qnr* and *tet*G genes at the end of the treatment. Yin et al. (2017) found that the multiple resistance genes and *intI*1 were significantly associated with the nitrate nitrogen during composting of dung manure. In addition, OM had positive correlations with most of the resistance genes (except *tet*G), indicating that the genes were decreased with the stabilization of sludge by earthworms.

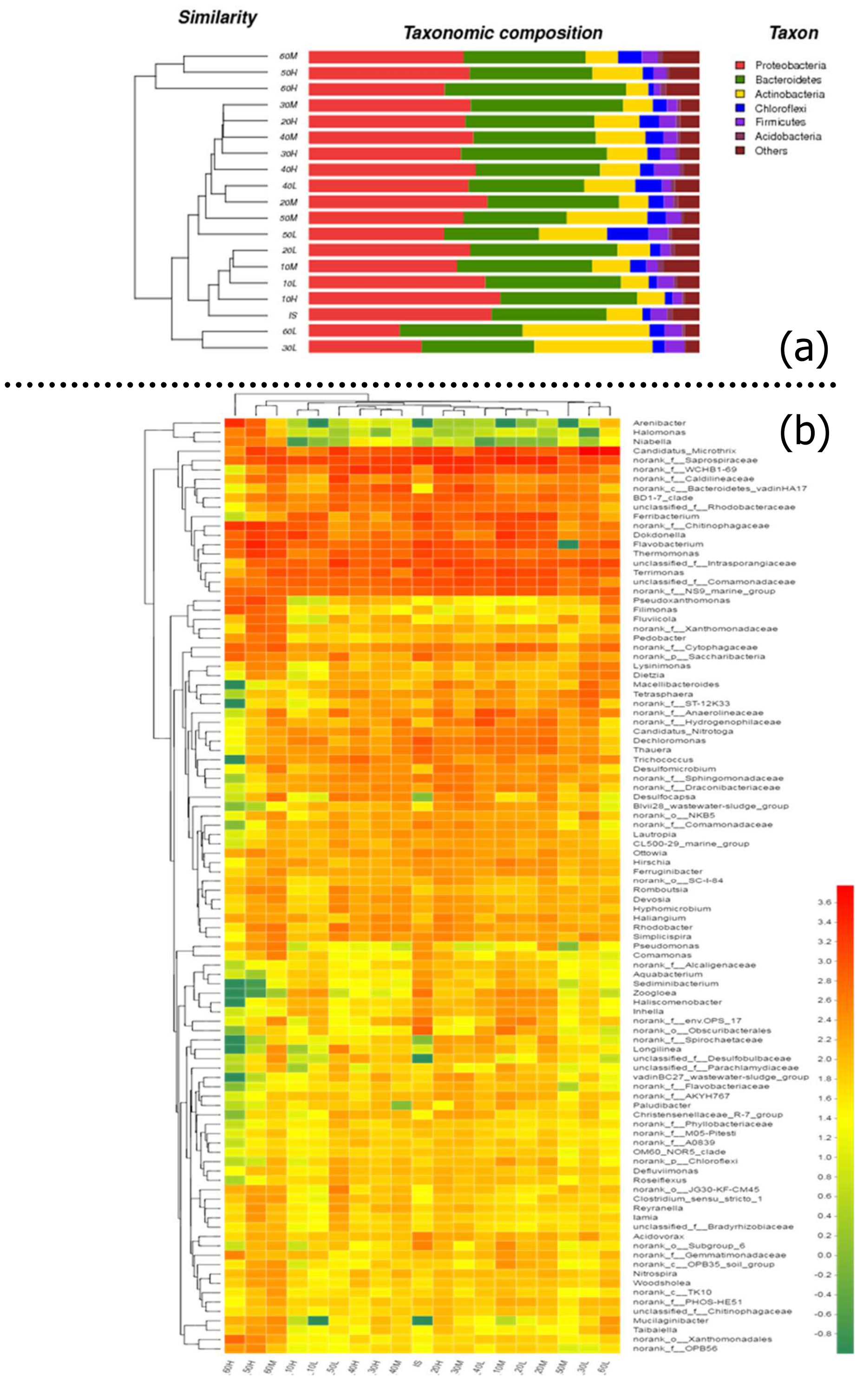

Figure 4 Variations of bacterial community composition in phylum level (a) and genus level (b) during vermicomposting of domestic excess sludge. IS-initial sludge. L, M and H represent treatments at 15°C, 20°C and 25°C. The number represent

sampling time (day). The number at the front of L, M and H represent sampling time (day).

Correlation between resistance genes and bacteria community (genus level) under different temperature conditions were investigated by RDA analysis as displayed in **Figure 5**. The results showed that the first and second components explained 72.8% of the total selected variables. The remarkable discrepancy of the relationship between bacterial community and resistance genes was observed during vermicomposting of sludge under three different temperature conditions. At the beginning of vermicomposting (10 day), the scores of samples from three treatments were positive on the RDA1 axis, with an order of 15°C >20°C >25°C, however, the scores from 20 day to 60 day were almost negative. The final vermicompost at 25°C got a positive score on the RDA1 axis and the highest score on the RDA2 axis, implying that the rebound of the resistance genes occurred at 25°C, which was consistent with the increase in the abundance of the resistance genes *tet*G, *tet*M, *qnr*A and *qnr*S on day 60. The increase was probably attributed to a relatively higher abundance of *Aeromonas* and *Chitinophagaceae* at the end of vermicomposting. It has been reported that *Aeromonas hydrophila* was not only indigenous to the earthworm *E. fetida* (Toyota and Kimura, 2000), the genus *Aeromonas* also carried integrase gene and plasmid, and was confirmed to be an important host bacteria of quinolone resistance genes (Huddleston et al., 2006; Cattoir, 2008; Jacobs and Chenia, 2017). Besides, vermicomposting temperature close to the recorded optimal growth temperature (22°C-37°C) of the genus may be an important reason of the sudden increase in the abundance of these genes in the final product. Zhu et al. (2017) have pointed out that *Chitinophagaceae* was a major host strain of tetracycline resistance genes. Bernard et al. (2012) confirmed that Chitinolytic organisms were strongly affected by earthworm activity, hence the highest activity of earthworm at 25°C (data not shown) contributed to a significant increase in the abundance of the bacterium.

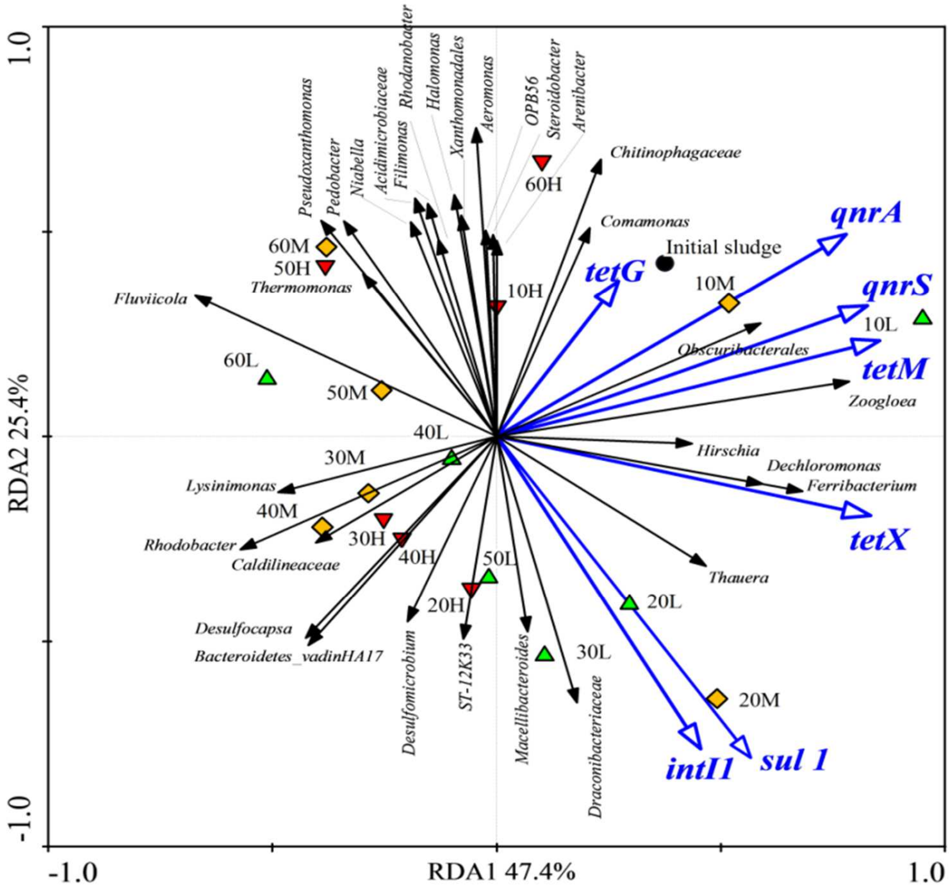

Figure 5 Redundancy analysis (RDA) of relationship between ARGs and bacterial community. IS-initial sludge. L, M and H represent treatments at 15°C, 20°C and 25°C. The number at the front of L, M and H represent sampling time (day).

To clarify possible host bacteria of resistance genes in the vermicomposting process, correlation analysis was carried out, as displayed in **Table 3**. *Ferribacterium* was a possible host bacterium of the resistance genes except for *tet*G, and *Zoogloa* and *Obscuribacvterale*s were possible host bacteria of resistance genes except for *sul*1 and *tet*G. *Thauera* may carry resistance genes like *sul*1, *tet*M and *tet*X. In addition, bacteria that can carry *intl*1 genes included *Dechloromonas*, *Ferribacterium*, *Macellibacteroides*, *Thauera*, *Draconibacteriaceae* and *ST-12K33*. These findings suggest that there are many different types of bacteria that were able to carry *intl1* gene in the excess sludge, which is consistent with the report by Huang et al. (2018). Possible host bacteria of *qnr*A and *qnr*S genes are mostly the same, except *Dechloromonas* that may only carry *qnr*A gene. The possible reason was attributed to different types of amino acid coding both resistance (Strahilevitz et al., 2009). Potential host bacteria of the three *tet* genes were quite different. Host bacteria were also affected by the type of resistance genes. In fact, this can be explained by the proven mechanism of tetracycline resistance, namely efflux pumps for *tet*G gene (Martinez et al., 2009), ribosome-protected proteins for *tet*M gene, encoding an enzyme that modified or inactivated tetracycline for *tet*X gene (Roberts, 2005; Sui et al., 2017).

Table 3 Pearson correlations between ARGs and bacterial community (genus level)

	*intl*1	*qnr*A	*qnr*S	*sul*1	*tet*G	*tet*M	*tet*X
Aeromonas	*-0.593			*-0.593			
Arenibacter	*-0.544						
Comamonas					*0.572		
Dechloromonas						*0.600	
Desulfocapsa		*-0.669				*-0.534	
Desulfomicrobium		*-0.540					
Ferribacterium						*0.625	*0.609
Filimonas	*-0.638						
Fluviicola	*-0.543			*-0.619		*-0.700	*-0.545
Halomonas	*-0.561						
Lysinimonas						*-0.606	
Macellibacteroides	*0.548						
Niabella	*-0.632						
Pedobacter	*-0.538			*-0.555			

Pseudoxanthomonas	*-0.655			*-0.567			
Rhodanobacter	*-0.614						
Rhodobacter		*-0.726				*-0.705	
Steroidobacter	*-0.555						
Thauera							*0.538
Thermomonas				*-0.544			
Zoogloea		*0.706	*0.622			*0.784	*0.640
Bacteroidetes_vadinHA17		*-0.591					
Acidimicrobiaceae	*-0.630						
Caldilineaceae		*-0.540				*-0.568	
Chitinophagaceae	*-0.559	*0.584					
Draconibacteriaceae	*0.619			**0.576			
OPB56	*-0.557						
ST-12K33	*0.526						
WCHB1-69		*-0.605					
Obscuribacterales		*0.657				*0.633	
Xanthomonadales	*-0.571						

Asterisk (*) represents significant difference at the level of $p < 0.01$. Values donate correlation (r^2). "-" means negative relationship between genus and ARGs.

Conclusions and perspectives

Vermicomposting at 25°C did not decrease ARGs of sludge in comparison to that at 15°C and 20°C. The attenuated resistance genes followed the first-order kinetic model, however, the magnitude of kinetic coefficients did not increase with the rise of temperature. The significant increment of nitrate nitrogen at 25°C was probably the main reason that contributed to the rebound of *qnr* and *tet*G genes at the end of vermicomposting. Vermicomposting of sewage sludge at 20 °C rather than 25 °C was recommended to obtain the final product with the efficiently attenuation of ARGs along with relatively high bio-stability.

Pathogenic bacteria (e.g. *Aeromonas* and *Xanthomonadales*) be capable of carrying with ARGs in sewage sludge were observed in the present study and the similar phenomenon was also confirmed by the previous study (Huang et al., 2020), indicating that ecological risk from sludge-derived bio-fertilizer generated by vermicomposting is non-negligible. Moreover, it is recommended that metatranscriptomics approach (Bashiardes et al., 2016), rather than conventional high throughput sequencing is applied to identify active bacterial community that is capable of carrying ARGs, in order to accurately evaluate occurrence status of active host bacteria in sewage sludge and clarify the involved mechanism of ARGs evolution during vermicomposting of sludge. It is primarily recognized that microplastics regardless from terrestrial or marine environment is an important vector for ARGs dissemination and their loaded ARGs in types and abundance obviously differed from the surrounding environments of microplastics (Liu et al., 2021). Importantly, microplastics and formed biofilm (termed as plasticsphere) could adsorb various contaminants such as heavy metals, pathogens and other emerging pollutants (Zhang and Chen, 2019; Amaral-Zettler et al., 2020; Liu et al., 2021), which may lead to more complicate mechanism of ARGs removal during vermicomposting.

CRediT authorship contribution statement

Guangyu Cui: Conceptualization, Methodology, Writing - review & editing. Xiaoyong Fu: Methodology, Validation. Sartaj Ahmad Bhat: Investigation, Writing-Review & Draft, Formal analysis. Weiping Tian: Methodology, Resources, Visualization. Xuyang Lei: Investigation, Resources. Yongfen Wei: Methodology, Resources, Visualization. Fusheng Li: Conceptualization, Visualization.

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